

1.Executive Summary

ABC E-Commerce is an online retail company operating across multiple regions. Management identified recurring inconsistencies in monthly sales reports, traced to underlying data quality issues in the source dataset of 5,000 sales transactions. This report documents the cleaning process applied to the raw data, presents the resulting sales performance analysis, and describes the interactive Excel dashboard built to support ongoing business decision-making.

2. Data Quality Issues Identified

Before analysis could begin, the raw transactional dataset was assessed and found to contain the following categories of data quality issues, consistent with common real-world e-commerce data problems:

Missing values — gaps in fields such as customer rating, delivery days, and payment method that would distort averages and category totals if left untreated.

Duplicate records — repeated transaction rows that would artificially inflate revenue, order counts, and category totals.

Inconsistent text formatting — category and region labels entered with mixed casing or extra whitespace (e.g., "home" appearing separately from "Home"), fragmenting totals that should sit under one category.

Invalid or out-of-range entries — values such as negative prices, impossible delivery days, or ratings outside the valid 1–5 scale.

3. Data Cleaning Methodology

The dataset was cleaned using Excel Power Query for validation and business user review. The process followed these steps: missing numeric values (delivery days, ratings) were imputed using medians or category-level averages, while missing categorical values were flagged or imputed using the mode; exact and near-duplicate transaction records were identified using order identifiers and key attribute combinations, then removed; category and region text was trimmed of whitespace and standardized to consistent title-case naming so that variants of the same label were correctly consolidated; and ratings, prices, and delivery days were checked against logical bounds, with invalid entries corrected or excluded.

4. Dashboard Results

The cleaned dataset was loaded into an Excel Pivot Table/Pivot Chart model to build an interactive dashboard with slicers for Product Category, Region, and timeline for Order Date, allowing all visuals to be filtered together. Core results are summarized below.

Key Performance Indicators

Total Revenue	\$5,104,379.37
Total Orders	20,201
Average Rating	2.97 / 5
Average Delivery Days	6.11 days

Revenue by Category

Category	Revenue (\$)
Electronics	1,829,297.34
Clothing	1,531,572.32
Home (combined)*	966,501.74
Beauty	761,425.79

Revenue by Region

Region	Revenue (\$)
West	1,345,561.03
North	1,281,508.45
South	1,242,205.81

Region	Revenue (\$)
East	1,235,104.08

Payment Method Distribution

Method	Share of Orders
Card	46%
Cash on Delivery	35%
Wallet	19%

5. Business Insights

Electronics is the highest revenue-generating category, indicating it should remain a priority for inventory investment and marketing spend.

The West region is the top-performing market, suggesting its regional strategies may be worth replicating elsewhere.

Card payments dominate customer purchases (46%), followed by COD (35%) and Wallet (19%).

There is an opportunity to grow digital wallet usage through cashback offers or discounts, which could also reduce the operational costs associated with COD.

Delivery times (6.11 days) are relatively long; combined with a low average rating (2.97/5), this points to fulfilment and customer experience as a priority area.

6. Recommendations

Investigate the revenue decline in the most recent period on the monthly trend chart to confirm whether it reflects a genuine trend or incomplete data.

Launch a digital wallet incentive campaign to shift volume away from COD, which carries higher processing and reconciliation costs.

Review logistics/courier performance to reduce delivery days, and investigate the drivers of the low average rating, as the two metrics are likely linked.

Continue prioritizing Electronics and Clothing given their outsized revenue contribution, while assessing whether Beauty and Home reflect genuine demand.

7. Conclusion

The cleaning process addressed missing values, duplicate records, inconsistent text formatting, and invalid entries in the ABC E-Commerce transactional dataset, producing a reliable base for analysis. The resulting dashboard gives management real-time, filterable visibility into revenue performance by category, region, payment method, and time period, and surfaces clear, actionable insights most notably the strength of Electronics and the West region, the dominance of card payments, and the need to improve delivery performance and customer satisfaction.